



JURONG JUNIOR COLLEGE

JC2 Preliminary Examination 2018

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

23 August 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

1	
2	
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11	
Total	

This document consists of **21** printed pages and **1** blank page.

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an electron micrograph of mitochondria cross-sections.

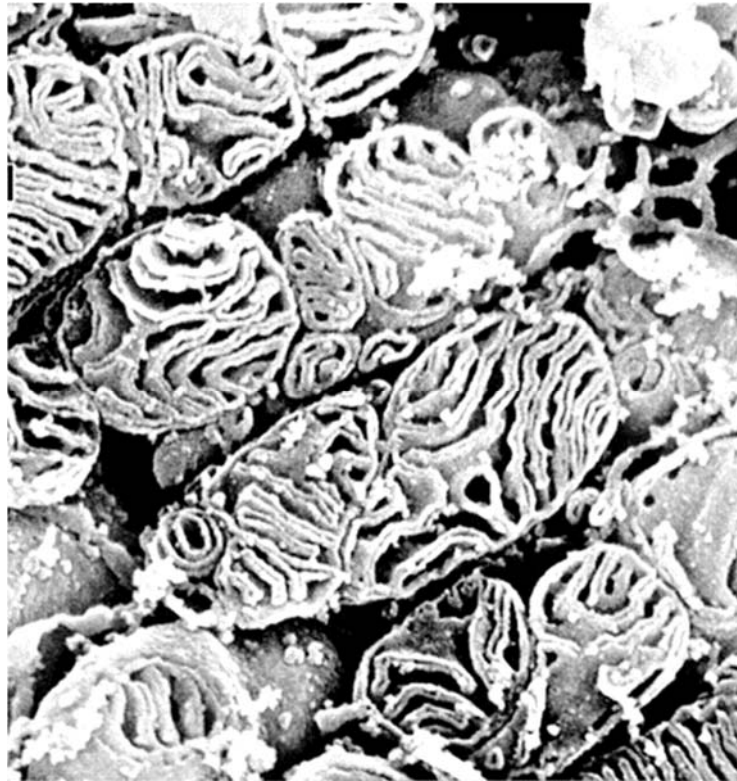


Fig. 1.1

- (a) With reference to Fig. 1.1, state one visible feature of the mitochondrion and explain how this feature is adapted for the mitochondrion's function. [3]

Visible feature:

1. The mitochondrion has a highly folded inner mitochondrial membrane / has numerous infoldings/cristae ;

Adaptation:

2. which provides a (large) surface (area) for the attachment of enzymes / electron transport chains / electron carriers / ATP synthase ; [R: for H^+ diffusion]
3. involved in oxidative phosphorylation for ATP synthesis. ;

OR

Visible feature:

4. The mitochondrion has a narrow intermembrane space / Compartmentalisation of the mitochondrion by the inner mitochondrial membrane into regions like the intermembrane space ;

Adaptation:

5. facilitates the accumulation / concentration of H^+ leading to the (rapid) establishing of a (steep) electrochemical / proton gradient required to ;
6. drive ATP synthesis. ;

[must make correct match between Visible Feature and Adaptation]

Fig. 1.2 shows the molecular structure of carbonyl cyanide-4-(trifluoromethoxy) phenylhydrazone (FCCP). FCCP is a respiratory poison that binds protons and transports them across the phospholipid bilayer of the inner mitochondrial membrane. Protons alone are unable to diffuse freely in this manner.

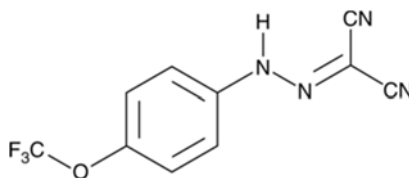


Fig. 1.2

(b) Explain, in relation to their properties, why FCCP readily diffuses across the phospholipid bilayer while protons do not. [2]

1. FCCP is a small, hydrophobic non-polar molecule and readily diffuses across the hydrophobic core of the phospholipids bilayer. FCCP is hence soluble in lipids. ;
2. Protons are charged (ions) and are repelled by the hydrophobic core of the phospholipid bilayer of the inner mitochondrial membrane. Thus, protons cannot diffuse readily. ;
[Mark once for “hydrophobic core”]

In the presence of FCCP, the rate of ATP synthesis during respiration is significantly diminished. The rate is further reduced if oxygen becomes unavailable. In such conditions where oxygen is depleted, cells continue to oxidise glucose to form pyruvate and sustain ATP synthesis.

(c) (i) Identify the key stages of aerobic respiration where the release of carbon dioxide occurs. [1]

- **Link reaction and Krebs cycle. ;**

(ii) State the location where the biochemical pathway enabling the continual oxidation of glucose to form pyruvate occurs. [1]

Cytosol / cytoplasm ;

(iii) Explain how ATP synthesis can be sustained in the absence of oxygen in yeast cells. [3]

1. **ATP in yeast can be sustained through anaerobic respiration, where glycolysis occurs followed by alcoholic fermentation. ;**
2. **Regeneration of NAD (during alcoholic fermentation) / Oxidation of NADH to NAD allows glycolysis to continue ;**
3. **resulting in a (net gain) of 2 ATP per molecule of glucose via substrate level phosphorylation. ;**

[Total: 10]

2 Fig. 2.1 shows a triglyceride molecule.

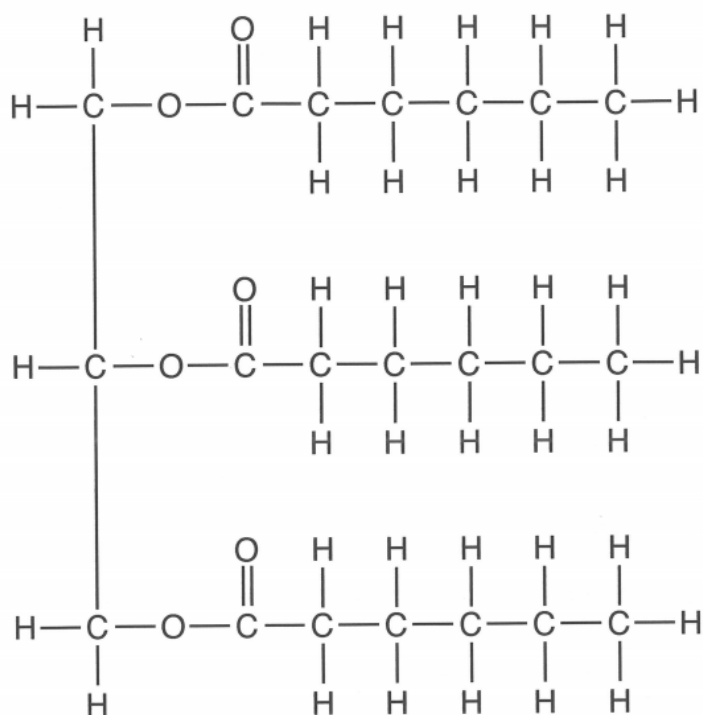


Fig. 2.1

(a) (i) State the names of the two types of molecules that undergo condensation reactions to form a triglyceride. [2]

1. **glycerol ;**
2. **fatty acid ;**

(ii) Describe what is meant by a condensation reaction. [2]

1. **A condensation reaction takes place between the -OH group and -COOH group ;**
2. **involves the removal of (one) water molecule ;**

A: Three condensation reaction and removal of three water molecules

(iii) The triglyceride in Fig. 2.1 is saturated.

Explain how the structure would be different for an unsaturated triglyceride. [3]

1. contains the lesser number of hydrogen atoms ;
2. presence of carbon-carbon double covalent bond ($C = C$) ;
3. presence of kinks in the hydrocarbon chain ;

(b) The eukaryotic cell surface membrane contains phospholipids, cholesterol and proteins.

(i) Describe how a phospholipid molecule differs from a triglyceride molecule. [2]

	Phospholipids	Triglycerides	
1.	1 glycerol, 2 fatty acids and 1 (-vely charged) phosphate group	1 glycerol and 3 fatty acids	;
2.	2 ester bonds and a phosphoester bond	3 ester bonds	;
3.	Phospholipid diversity is based on differences in the two fatty acids and in the groups attached to the phosphate group of the head.	Triglyceride diversity is based on differences in the three fatty acids.	;
4.	Phospholipids are amphipathic in nature.	Triglycerides are non-polar in nature.	;
5.	Phospholipids may associate covalently with carbohydrates to form glycolipids.	Triglycerides do not associate covalently with carbohydrates.	;

* A: fatty acid tails / hydrocarbon chains.

(ii) Describe the roles of cholesterol in eukaryotic cell surface membranes. [2]

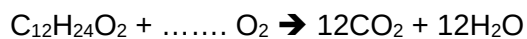
1. increases the stability of membranes/decrease membrane fluidity, by restraining the movement of phospholipids, at (relatively) warm temperatures ;
or
2. increases the flexibility of membranes/increase membrane fluidity, by hindering the close packing of phospholipids, at low temperatures ;
I: helps to reduce the tendency of membrane to 'freeze'/solidfy/breaking up (consequence)
3. acts like a plug, reducing the escape / entry of charged ions/small polar molecules through the membrane ;

- (c) The respiratory quotient, RQ, is used to show which substrate is being metabolised by cells. It can be determined using the equation below.

$$\text{RQ} = \frac{\text{molecules of carbon dioxide released}}{\text{molecules of oxygen taken in}}$$

Lauric acid is a saturated fatty acid found in coconuts and has a chain of 12 carbon atoms.

- (i) Complete the equation below which outlines the aerobic respiration of lauric acid. [1]



- 17 ;

- (ii) Calculate the RQ value for lauric acid.

Give your answer to 2 decimal places.

- allow ecf from (i) for one mark

RQ value = **0.71** [1]

[Total: 13]

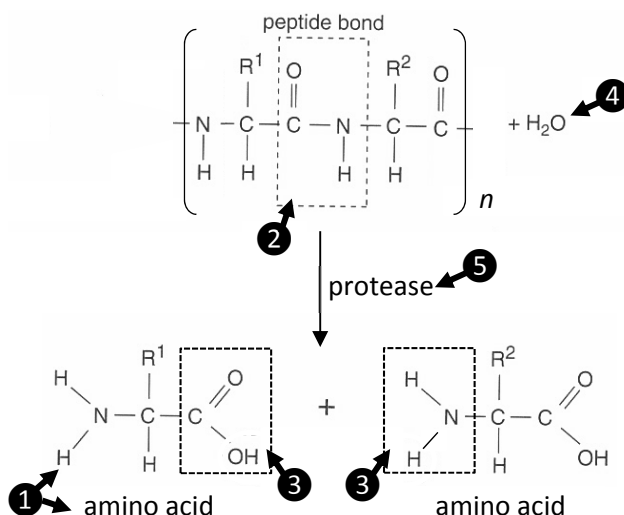
3 Lysosomes are important membrane-bound organelles within animal cells.

They contain many different types of enzymes that hydrolyse lipids, nucleic acids, polysaccharides or proteins. When phagocytic vesicles containing these food substances fuse with lysosomes, the internal pH of lysosomes will be lowered to a range of pH 4.5 to pH 5.0 resulting in the activation of the lysosome enzymes.

(a) Describe how proteins are hydrolysed.

You may use annotated diagrams to illustrate your answer. [4]

1. Hydrolysis of proteins / polypeptides to form dipeptides / amino acids / short peptides ;
 2. involves the breaking of peptide bonds along a polypeptide chain ;
 3. to form amino ($-\text{NH}_2$) groups and carboxyl ($-\text{COOH}$) groups on adjacent amino acid residues. ;
 4. The reaction involves the addition of water / hydrolysis ;
 5. and is catalysed by proteases / peptidases. ;
- [max 4m]



Award for corresponding points in diagram:

1. Structure of amino acid drawn correctly along with label for “amino acid”. Ignore errors for carboxyl and amino group (penalise in Pt 3).
2. Peptide bond drawn correctly + to show conversion of polypeptide (A: if dipeptide is shown) to smaller molecules (amino acids)
3. *Both* carboxyl group and amino group drawn correctly for *both* molecules of amino acids.
4. Addition of water [Ignore number of water molecules]
5. Example of valid type of enzyme (i.e. must be a protease)

(b) Explain how the lysosome enzymes can be activated by the change in pH. [3]

1. Changes in pH alters the ionic charges on acidic and basic R-groups of amino acid residues present in inactive lysosome enzymes. ;
2. This results in the rearrangement of ionic bonds and hydrogen bonds in lysosome enzymes and ;
A: break AND form new ionic bonds
3. alters the 3D conformation of the active sites of these enzymes such that these active sites are now complementary to the shape of their substrates / substrates can enter and bind to the active site / AW. ;

(c) Suggest why the lysosome membrane is not destroyed by the enzymes in the lysosome. [3]

1. The lysosome membrane contains glycoproteins / glycolipids / phospholipids;
2. Lysosomes contains only enzymes that can breakdown lipids / nucleic acids / proteins / polysaccharides ;
3. The enzymes in the lysosome are specific (to the lipids / nucleic acids / proteins / polysaccharides in the phagocytic vesicle). ;
 OR
4. This is because the shape of these enzymes' active sites are not complementary to the membrane glycoproteins / glycolipids / phospholipids. Hence, membrane glycoproteins / glycolipids / phospholipids are unable to enter and bind to the active sites of the enzymes to form substrate-enzyme complexes for hydrolysis to occur / AW (ref. to lack of fit between substrate and active site of enzyme) ;
5. and hence are unable to digest the glycoproteins / glycolipids / phospholipids in the lysosome membrane. ;
 Hence the integrity of the lysosome membrane is preserved.

[Total: 10]

4 Fig. 4.1 shows the elongation phase of translation.

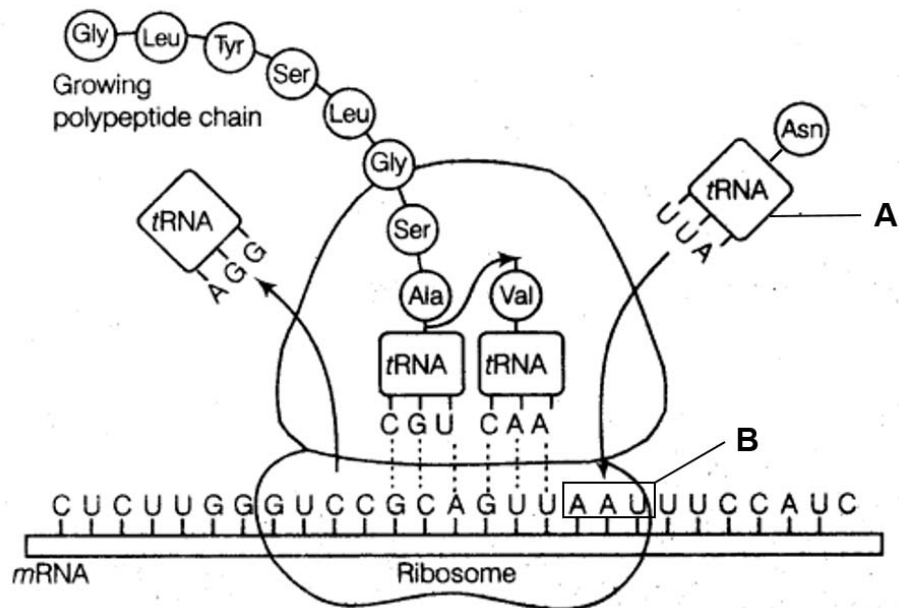


Fig. 4.1

(a) On Fig. 4.1, draw an arrow to show the direction of translocation of the ribosome. [1]

(b) Name the structures **A** and **B** in Fig. 4.1. [2]

A: aminoacyl-tRNA ;
B: codon ;

(c) Explain how the molecular structure of **A** is related to its functions. [2]

Structure

1. Has anticodon (loop) that complementary base pairs with a particular codon on the mRNA
2. Attached covalently to its specific amino acid coded for by the anticodon of the tRNA at the CCA stem

Function (OWTTE)

3. To transfer amino acids present in the cytoplasm to the ribosome ;
4. To act as an intermediate molecule between the codon of mRNA and the amino acid sequence of the polypeptide strand ;

1+4 / 2+3 / 2+4

(d) Describe the phase of translation that occurs before elongation. [3]

1. A small ribosomal subunit recognises and binds to the 5' end of the mRNA and travels along the mRNA until it reaches the first AUG / start codon ;
2. A special initiator tRNA carrying the amino acid methionine (Met) / anticodon UAC, binds to the start codon AUG on the mRNA ;
3. The union of mRNA, initiator tRNA, and a small ribosomal subunit is followed by the attachment of a large ribosomal subunit, completing a translation initiation complex ;
4. Proteins called initiation factors and GTP are required to bring all these components together ;

(e) State two differences between translation in prokaryotes and translation in eukaryotes. [2]

	Features	prokaryotes	eukaryotes	
1.	Ribosomes involved	Involves 70S ribosomes	Involves 80S ribosomes	;
2.	Continuous/ simultaneous transcription and translation	Translation is continuous/ simultaneous with transcription	Translation is separated/ not simultaneous with transcription	;
3.	mRNA involved in translation	mRNA used is usually polycistronic/code for many proteins/ act as template for the synthesis of many polypeptides	mRNA used is monocistronic/code for one protein/ act as template for the synthesis of one polypeptide;	;
4.	Location	Translation occurs on free ribosomes only	Translation occurs on free ribosomes or ribosomes attached to endoplasmic reticulum	;

[Total: 10]

- 5** Viruses such as bacteriophages have been described as being “organisms at the edge of life”.

(a) State one reason why viruses may be classified as:

(i) living [1]

- 1. They are capable of reproduction only in living host cells ;**
- 2. They are capable of transmitting/passing on genetic characteristics from one generation of viruses to the next generation of viruses ;**
- 3. Genomes of viruses are capable of mutating ;**

(ii) non-living. [1]

- 1. They are acellular ;**
- 2. They cannot carry out metabolism on their own and must replicate using the host cell's metabolic machinery/enzymes and ribosomes ;**
- 3. They do not grow or divide/ new viral components are synthesised and assembled within the infected host cell ;**
- 4. They contain only one type of hereditary material i.e. either DNA or RNA but never both ;**
- 5. They survive within a host cell but crystallise when outside of host cell ;**
- 6. They are not capable of movement as they have no flagella or other means to propel themselves ;**

AVP: unable to obtain own source of nutrients through diet or photosynthesis etc.

(b) Fig. 5.1 is an electron micrograph of a T4 bacteriophage.

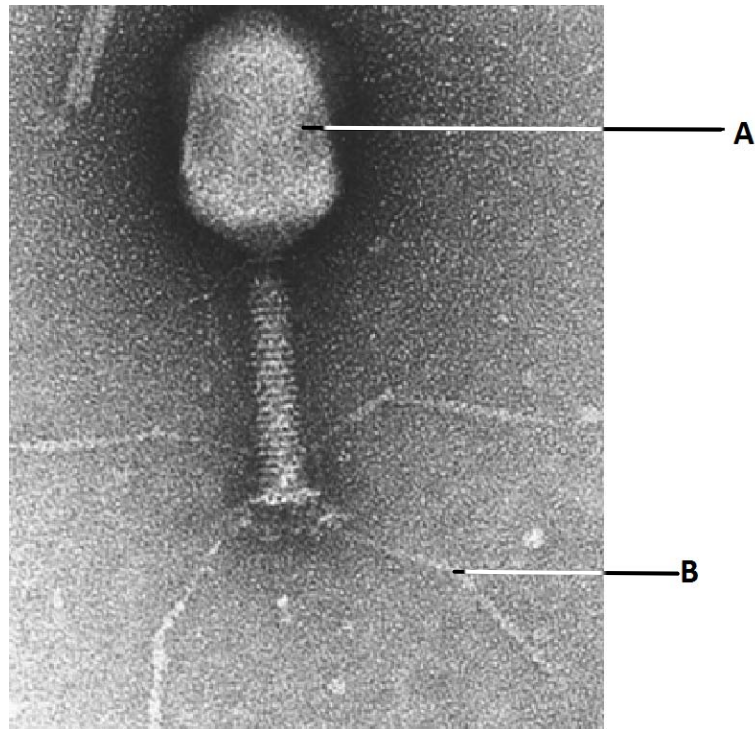


Fig. 5.1

(i) Explain what is meant by the term bacteriophage. [1]

1. Viruses that only infect bacteria ;

(ii) Identify the structures labelled **A** and **B**. [2]

A: capsid ;

B: tail fibre ;

Fig. 5.2 shows a scanning electron micrograph of numerous T4 bacteriophages infecting *Escherichia coli*.



Fig. 5.2

(c) With reference to Fig. 5.2, outline the process which allows for *E.coli* DNA to be transferred to another bacterial cell after infection. [4]

1. **Generalised transduction** ;
2. T4 bacteriophages undergo lytic cycle /phage enzymes (nucleases) hydrolyse the bacterial chromosome into fragments ;
3. A fragment of the bacterial/host cell's degraded DNA (instead of phage genome) gets mistakenly packaged into the capsid during assembly of the phage ;
4. The defective phage attaches to another bacterium and injects the piece of bacterial DNA acquired from the lysed bacterial cell ;
5. Foreign bacterial DNA incorporated into new host cell's DNA via homologous recombination (i.e. crossing over with a homologous region found on the recipient bacterial chromosome) ;

[Total: 9]

- 6 In *Escherichia coli*, the level of transcription for most genes varies widely according to the nutrient growth condition. The cause of these differences may be due to the presence of many operon-specific activators or repressors, which vary with the composition of the growth medium.

Fig. 6.1 shows the interaction between a lac repressor, coded by a regulatory gene, and the *lac* operon.

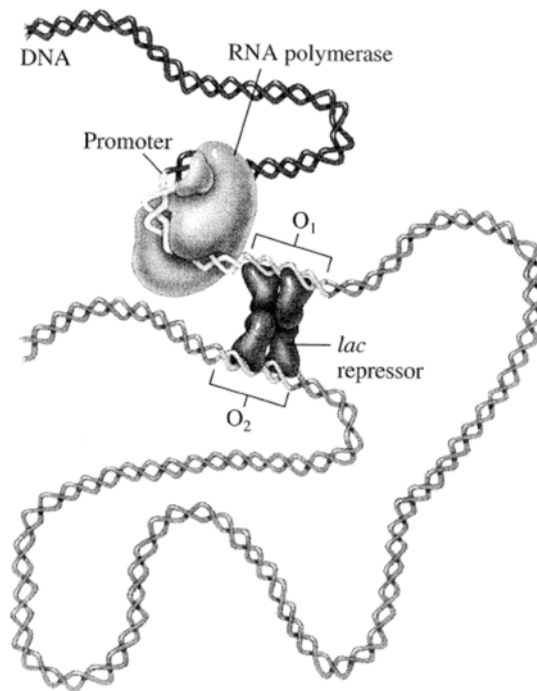


Fig. 6.1

(a) Explain what is meant by *regulatory gene*. [1]

1. A regulatory gene codes for a specific protein product that regulates the expression of the structural genes ;

(b) Name two enzymes coded by structural genes of the *lac* operon. [2]

1. β -galactosidase ;
2. lac permease ;
3. β -galactoside transacetylase ;
(any 2)

(c) With reference to Fig. 6.1,

(i) suggest the nutrient growth condition of the *E. coli* [1]

1. Absence of lactose in the growth media/environment ;

(ii) outline how the lac repressor interacts with the *lac* operon. [2]

1. lac repressor is synthesised in its active conformation and recognises and binds to the operator ;
2. No allolactose binds to the allosteric site of lac repressor, lac repressor remains in its active 3D conformation and remains bound to the operator ;

Fig. 6.2 shows the *trp* operon of *E. coli* with five genes *trpA*, *trpB*, *trpC*, *trpD* and *trpE* that code for enzymes responsible for the biosynthesis of the amino acid tryptophan (Trp).

This operon is transcriptionally regulated by *trp* repressor, encoded by the *trpR* gene, which is located upstream of *trp* operon.

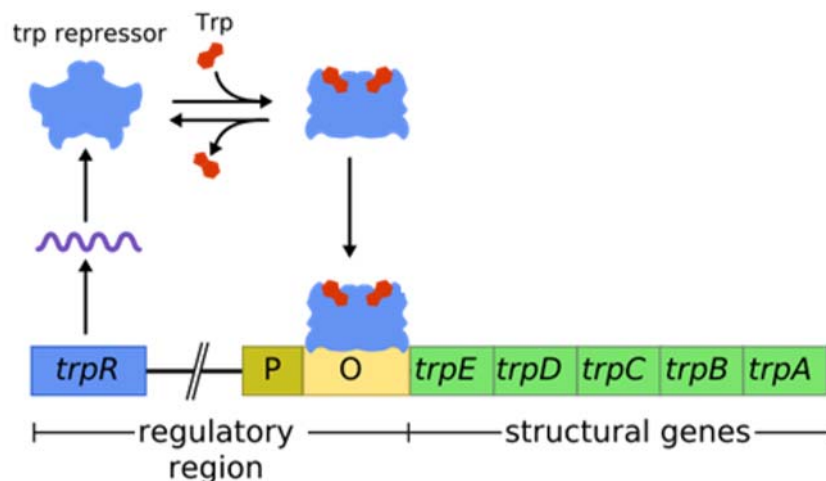


Fig. 6.2

(d) *trp* operon is an example of repressible operon while *lac* operon is an example of inducible operon.

State two other differences between *trp* operon and *lac* operon. [2]

	features	<i>trp</i> operon	<i>lac</i> operon	
1.	Effector molecule	tryptophan acts as a co-repressor	Allolactose acts as an inducer	;
2.	Effect of effector molecule on operon	In the presence of co-repressor/tryptophan, operon is turned off	In the presence of inducer/allolactose, operon is turned on	;
3.	Conformation of newly synthesized repressor	Repressor is synthesized in the <u>inactive form</u>	Repressor is synthesized in the <u>active form</u>	;
4.	Nature of pathway involved	<i>trp</i> operon is involved in <u>anabolic pathway</u>	<i>lac</i> operon is involved in <u>catabolic pathway</u>	;
5.	By default, operon are switched on/off	operon are switched on	operon are switched off	;
6.	When does repressor bind operator	When <u>co-repressor binds to allosteric site</u> , repressor can recognize and bind the operator	Repressor in <u>its active form</u> , and recognize and bind to the operator	;
7.	When does repressor not bind to operator	Repressor in <u>its inactive form</u> , and does not bind to the operator	When <u>inducer binds to allosteric site</u> , repressor does not bind to the operator	

[Total: 8]

7 Fig. 7.1 shows an electron micrograph of a chromosome in prophase II.

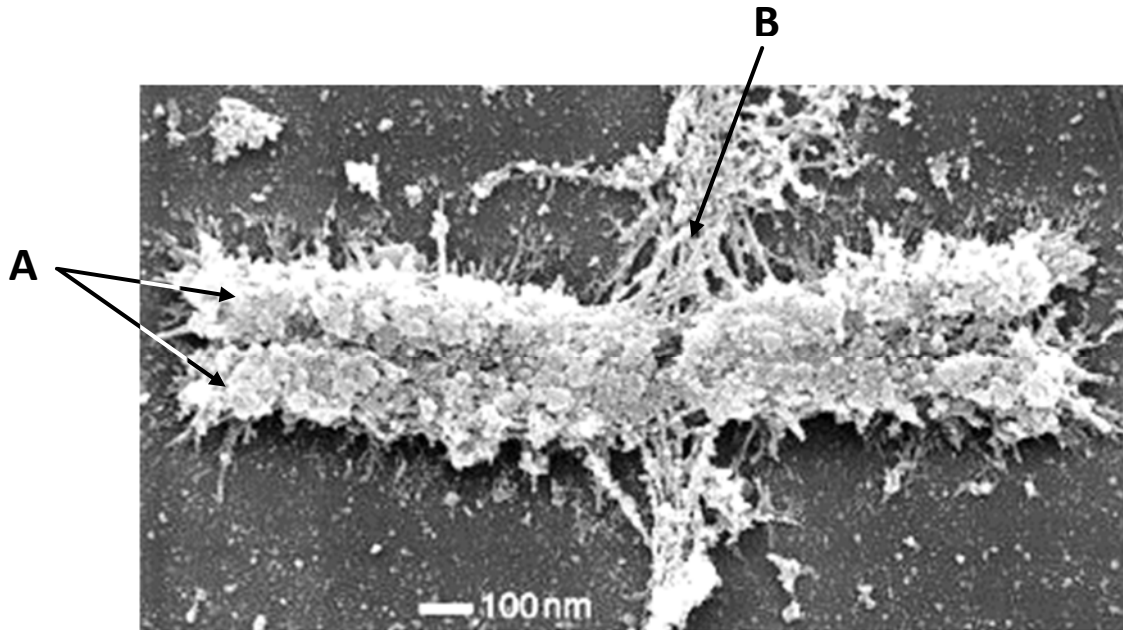


Fig. 7.1

(a) Name the structures **A** and **B**. [2]

A: (a pair of recombinant) chromatids ; [R: sister chromatids / chromosomes]

B: (kinetochore) spindle fibre ;

(b) Explain why the chromosome occurs as a double structure. [2]

1. Semi-conservative DNA replication occurred during S phase of interphase, producing two identical DNA molecules ;
2. which coil and condense during prophase I of meiosis I to form a chromosome consisting of 2 (identical) sister chromatids held together at the centromere before crossing over occurred. ;

The risk of mis-segregation of chromosomes increases with age among women. This can lead to aneuploid embryos. Fig. 7.2 shows an oocyte undergoing nuclear division with mis-segregated and lagging chromosomes.

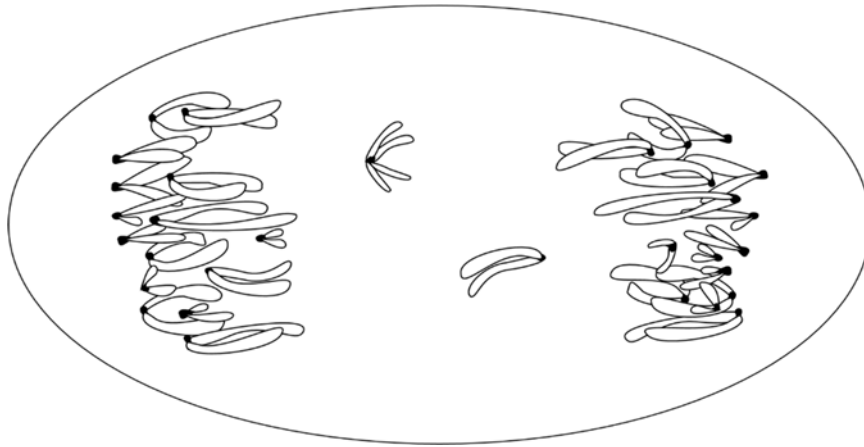


Fig. 7.2

(c) Identify the stage in meiosis as shown in Fig. 7.2. [1]

Anaphase II ;

(d) With reference to Fig. 7.2, suggest how aneuploid embryos are formed. [3]

1. **Mis-segregation of chromosomes arose when centromeres did not divide properly / recombinant chromatids did not separate at their centromeres. ;**
OR
2. **Lagging chromosomes arose when the kinetochore spindle fibres attached to the centromeres failed to shorten and contract to pull the daughter recombinant chromosomes to opposite poles of the cell. ;**
3. **These processes may (lead to chromosomal/numerical aberrations) resulting in oocytes with extra / less / $n + 1$ / $n - 1$ chromosomes. [No credit for mentioning “chromosomal/numerical aberrations” alone];**
4. **Fusion of these oocytes with a normal haploid sperm, results in aneuploid embryos. ;**

[Total: 8]

- 8 The sweet pea, *Lathyrus odoratus*, is a flowering plant that grows in many parts of Europe. The inheritance of flower colour and shape of pollen grains is controlled by genes that display autosomal linkage.

(a) Explain what is meant by autosomal linkage. [2]

1. **two or more genes on same chromosome will be inherited together;**
AND
2. **not sex chromosome/non-sex chromosome;**

(b) In sweet peas, the allele for purple flowers is dominant to the allele for red flowers and the allele for long pollen grains is dominant to the allele for round pollen grains.

A sweet pea plant that is heterozygous for both purple flowers and long pollen grains is crossed with a sweet pea plant with red flowers and round pollen grains. The results of this cross are shown in Table 8.1.

Table 8.1

offspring phenotype	number of offspring
purple flowers, long pollen grains	44
red flowers, round pollen grains	44
purple flowers, round pollen grains	6
red flowers, long pollen grains	6

Describe and explain the results shown in Table 8.1. [3]

1. The observed offspring of the cross are **minority of recombinants combination/phenotype**;
2. and **majority of parental combinations/phenotype** / The 2 genes for flower colour and shape of pollen grains are linked and the alleles of the two genes will be inherited together as one linkage group, resulting in a **higher proportion of gametes/ higher chances of getting gametes carrying the parental types/parental gametes** ;
3. **Crossing over between the two linked genes** on non-sister chromatids of homologous chromosomes may occur, (as crossing over is a chance event,) resulting in a lower proportion of **recombinant gametes**/lower chances of getting recombinant gametes ;

(c) Draw a genetic diagram to explain the results shown in Table 8.1. [3]

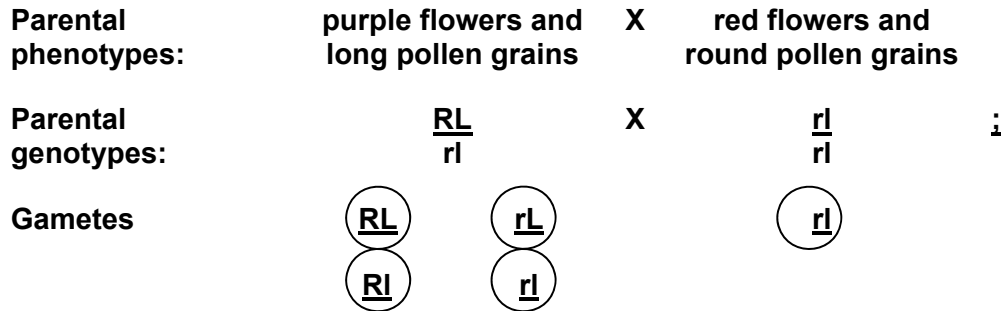
Use the following symbols to represent the different alleles involved:

R - purple flower

r - red flower

L - long pollen grain

l - round pollen grain



Fertilisation:

	$\frac{RL}{}$	$\frac{rl}{}$	$\frac{Rl}{}$	$\frac{rL}{}$
$\frac{rl}{}$	$\frac{RL}{rl}$	$\frac{rl}{rl}$	$\frac{Rl}{rl}$	$\frac{rL}{rl}$

Offspring genotype :	$\frac{RL}{rl}$	:	$\frac{rl}{rl}$	:	$\frac{Rl}{rl}$	:	$\frac{rL}{rl}$	;
Offspring phenotype :	purple flowers, long pollen grains	:	red flowers, round pollen grains	:	purple flowers, round pollen grains	:	red flowers, long pollen grains	;
Offspring phenotypic ratio :	7	:	7	:	1	:	1	

A horticulturist suggested the hypothesis that the phenotypic ratio of the offspring from the cross was 1:1:1:1. A chi-squared test was carried out on the results of the cross.

phenotype	observed number (O)	expected number (E)	$\frac{(O - E)^2}{E}$
purple flowers, long pollen grains	44	25	14.44
red flowers, round pollen grains	44	25	14.44
purple flowers, round pollen grains	6	25	14.44
red flowers, long pollen grains	6	25	14.44
			$\chi^2 = 57.8$

Table 8.2 shows part of the critical values of the chi-squared distribution.

Table 8.2

df	Probability								
	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750

(d) Explain the significance of the chi-squared value for these results. [3]

1. The calculated χ^2 value is 57.8, more than the critical χ^2 value of 7.815 at $p = 0.05$, Value of p is less than 0.005, less than $p = 0.05$;
2. Reject the null hypothesis, there is significant difference between the observed and the expected ratio ;
A: results/data
3. The phenotypic ratio of the offspring from the cross was not 1:1:1:1, any deviation from the expected is not due to chance but due to other factors eg. Linkage ;

[Total: 11]

- 9 On the volcanic, equatorial island of São Tomé found off the coast of western Africa, two species of vinegar flies *Drosophila yakuba* and *Drosophila santomea* live in co-existence. While *D. santomea* is found only in the moist forests at higher elevations, its close relative, *D. yakuba* resides mostly in the drier lowlands disturbed by human activities. *D. santomea* is also found exclusively on São Tomé while *D. yakuba* can also be found on the neighbouring islands of Príncipe and Bioko and on continental Africa.

Fig. 9.1 shows the distribution patterns of the two species of vinegar flies on São Tomé.

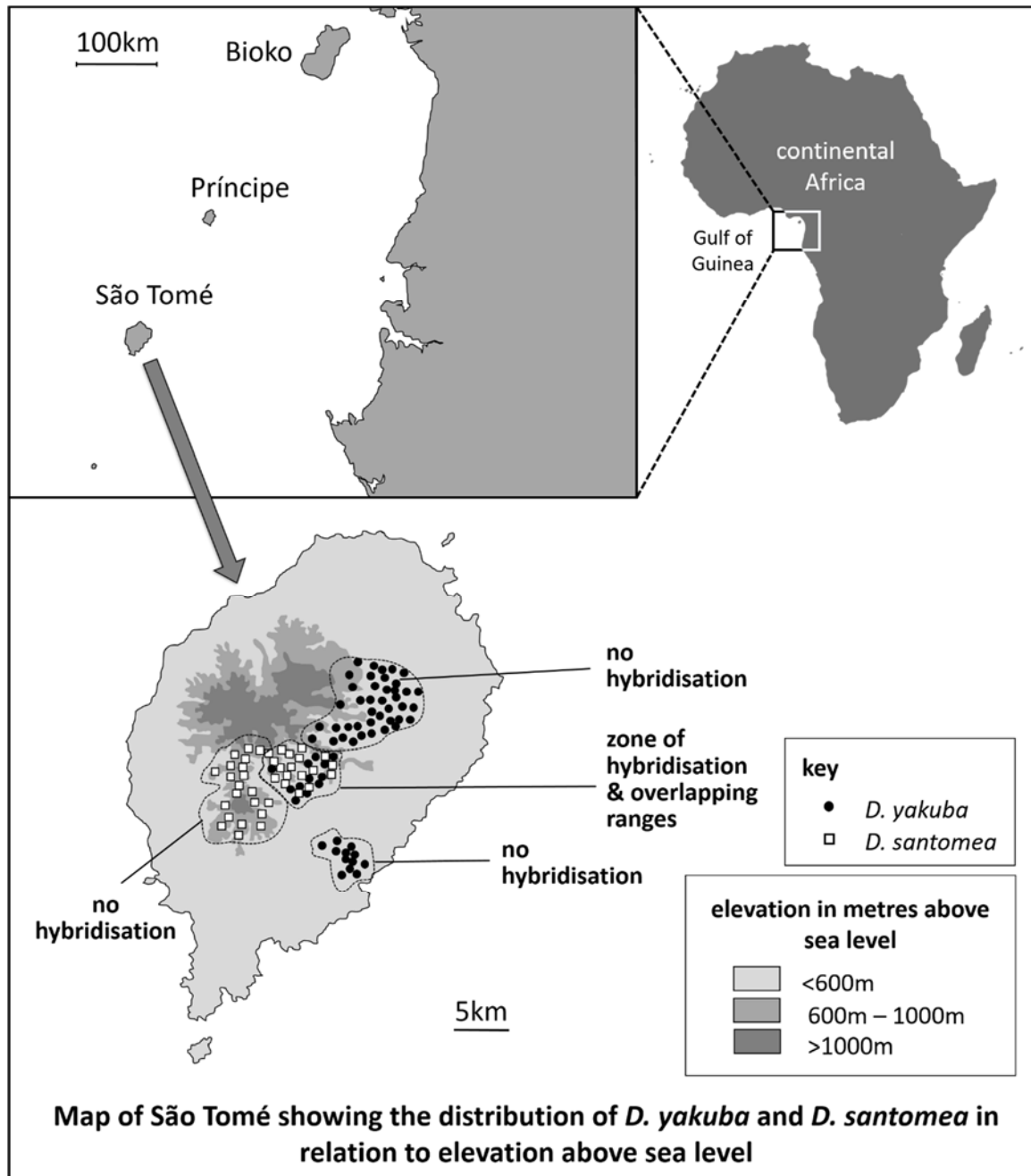


Fig. 9.1

(a) Give one reason why *D. yakuba* found on São Tomé and continental Africa might be considered as the same species. [1]

1. These populations of *D. yakuba* are capable of interbreeding to produce viable and fertile offspring ;
2. And share the same ecological niche ; [R: same environment]
3. And share similar body shape and/or other structural features / morphology ; [max 1m]

(b) With reference to Fig. 9.1, explain how island species like *D. santomea* could have arisen from an ancestral population of vinegar flies. [5]

1. Founder effect occurs when some members of an ancestral vinegar fly population from continental Africa colonised the islands along the coast to form sub-populations ; [Reject if wrong reference to migration from lower elevation to higher elevation on same island was mentioned]
2. The sub-populations become geographically isolated leading to allopatric speciation (and are also physiologically / behaviourally isolated leading to sympatric speciation) ;
3. The sub-populations did not interbreed (when isolated) and thus gene flow was disrupted ;
4. The sub-populations were exposed to different environments on each island and were thus subjected to different selection pressures e.g. availability of food, predators etc. ;
5. Since there was variation within the sub-populations, individuals with favourable characteristics were at a selective advantage (or vice versa) and
6. can survive to maturity, reproduce and passed on their favourable alleles / genes (R: traits) to their offspring (vice versa) ;
7. Over successive generations, evolutionary changes occurred independently in each subpopulation / changes in allele frequencies in each sub-populations lead to divergence, giving rise to a new species such as *D. santomea*. ;

It was observed that *D. yakuba* and *D. santomea* can hybridise in a well-delineated zone of hybridisation. Hybrids possess intermediate traits of their parents. A population count revealed that hybrids were outnumbered by both *D. yakuba* and *D. santomea* in this zone.

(c) Suggest why the hybrid population is smaller than *D. yakuba* and *D. santomea* populations in this zone. [1]

1. Hybrids have lower fitness / are selectively less advantageous than *D. yakuba* and *D. santomea* / disruptive selection against (the intermediate traits of) hybrids / ORA. ; [A: “hybrids are less adapted to the environment / have less favourable traits (R: alleles) compared to pure breeds”]
2. Interspecific breeding (breeding between *D. yakuba* and *D. santomea*) occurs less frequently than intraspecific breeding (within each species). ;
3. Hybrids have lower reproductive potential compared to *D. yakuba* and *D. santomea* pure breeds. ; [R: hybrids are infertile and cannot produce viable and fertile offspring]

R:

- *D. yakuba* and *D. santomea* have low chance of encountering each other.
- Behavioural and physiological isolation occurs between *D. yakuba* and *D. santomea*.

Studies of the nuclear genome suggested that *D. yakuba* on São Tomé is more closely related to *D. yakuba* from continental Africa than to *D. santomea*. Fig. 9.2 shows the evolutionary relationships among these populations and their sister species *D. teissieri*.



Fig. 9.2

Based on this data, biologists argued that there were two occasions where ancestral populations of vinegar flies arrived on São Tomé from continental Africa and rapidly colonised parts of the island.

(d) Using Fig. 9.2, explain how biologists came to the conclusion that two colonisation events had taken place on São Tomé. [2]

1. Based on the phylogenetic tree, São Tomé *D. yakuba* population shared a most recent common ancestor with the *D. yakuba* population from continental Africa, ;
2. suggesting that the São Tomé *D. yakuba* population had only recently arrived on the island from continental Africa ;
[idea of second recent dispersal and colonisation of *D. yakuba*; must match correct colonisation event to correct species]
- OR
3. where an existing *D. santomea* population had descended from an ancestral population from an initial colonisation event. ;
[idea of first dispersal and colonisation of *D. santomea*; must match correct colonisation event to correct species]

[Pt 1, 2 and 3 must identify correct clades: e.g. Pt 1 must mention the clade that only includes São Tomé *D. yakuba*, continental African *D. yakuba* and their common ancestor and excludes *D. santomea*]

[Total: 9]

10 Phagocytes and lymphocytes are both involved in defence against infectious diseases.

Active B lymphocytes are known as plasma cells.

Fig. 10.1 shows drawings made from electron micrographs of a phagocyte, **A**, and a plasma cell, **B**.

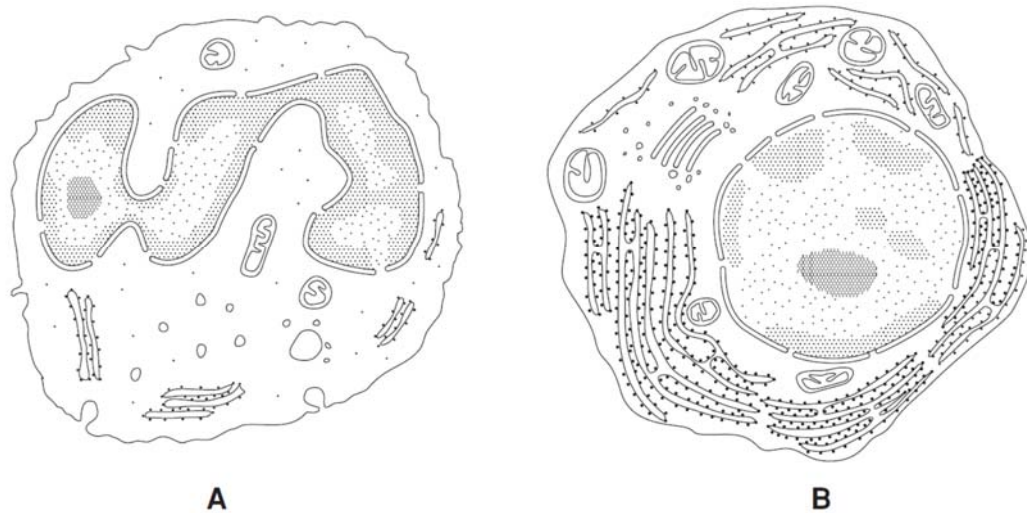


Fig. 10.1

(a) State two visible structural differences between the cells **A** and **B**. [2]

Structural feature	Phagocyte (A)	Plasma cell (B)
1. Rough endoplasmic reticulum (RER)	Small quantity	Large quantity
2. Ribosomes	Few OR Free ribosomes	Many OR RER bound ribosomes
3. Lysosomes/vacuoles	Present	Absent
4. Nucleus	Lobed/irregular/not round	Round/not lobed/regular/ circular
5. Golgi body	Absent	Present
6. Mitochondria	Few	Many
7. Presence of endocytotic/pinocytotic/ phagocytic/exocytotic vesicles	Present	Absent

(b) With reference to Fig. 10.1, describe the modes of action of the two cells in defence against infectious diseases. [4]

phagocyte

1. ingest/engulf/phagocytosis of / endocytosis of, bacteria/microbes/pathogens ;
R: antigens, virus
2. (form) phagocytic / endocytotic, vacuoles/vesicles/phagosomes ;
3. lysosomes contains hydrolytic enzymes to digest / hydrolyse, (bacteria / AW) ;
4. antigen presentation;

max 2

plasma cell

5. produce / secrete / release / synthesise , antibodies ;
6. antibodies are proteins synthesized by ribosomes on RER ;
7. Golgi (body) packages antibodies ;
8. each type of plasma cell produces one type of antibody ;
9. antibody effects agglutination/opsonisation/neutralisation ;

max 2

[Total: 6]

- 11 An analysis of ice cores from the Arctic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 11.1 shows the concentrations of carbon dioxide measured in ice cores, dated between 1000 and 2000 AD.

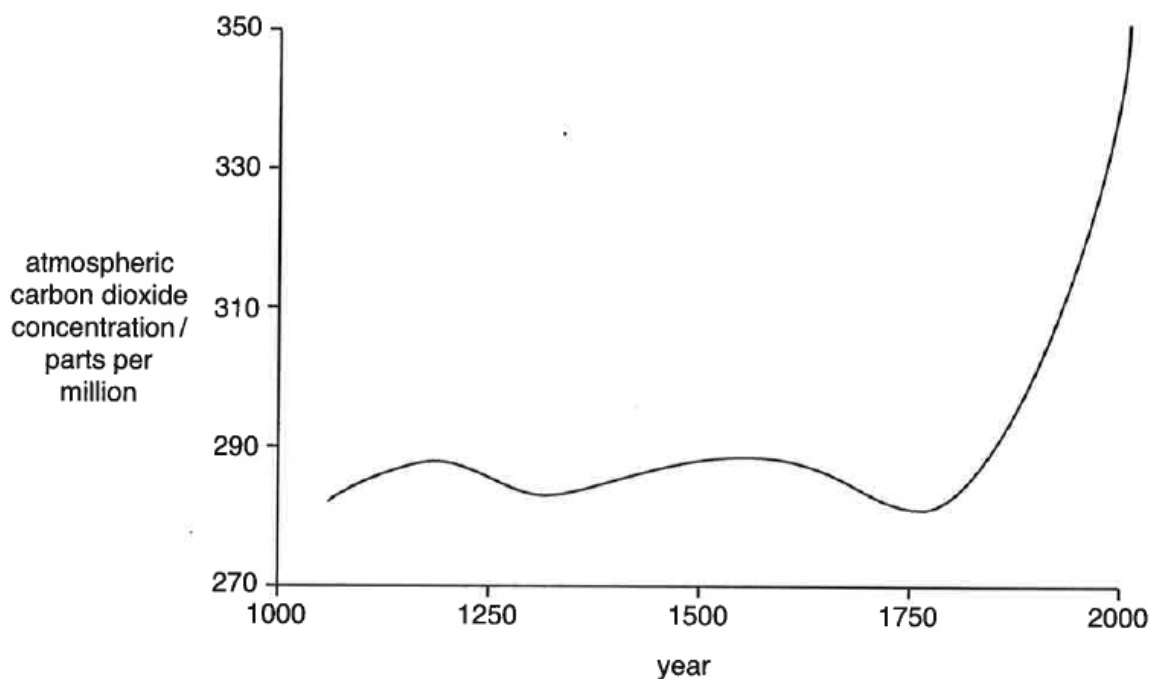


Fig. 11.1

(a) Describe the trend in Fig. 11.1. [2]

- The atmospheric carbon dioxide concentration stayed relatively constant at around 285 parts per million in year 1060 and increased exponentially/sharply in year 1750 to reach 350 parts per million in year 2000 ;

1m for constant, increased sharply
1m for pair of figures

(b) Atmospheric carbon dioxide concentrations show regular annual variations. Suggest one reason for this. [1]

- There are (four) seasons/seasonal changes in a year, atmospheric carbon dioxide concentrations is lower in summer and higher in winter ;

- (c) Fig. 11.2 shows that, over the same period of time, the average surface temperature of the Earth has shown a similar pattern of change. The increasing concentrations of carbon dioxide is thought to be responsible for the increase in temperature over the last 100 years. This is referred to as the enhanced greenhouse effect.

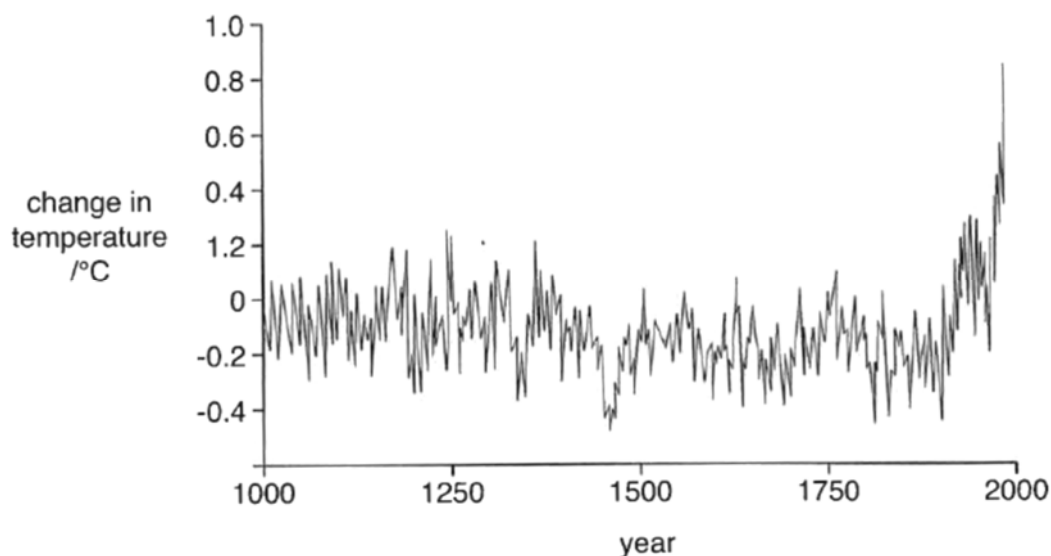


Fig. 11.2

- (i) Describe one way in which the data in Fig. 11.2 resembles the data in Fig. 11.1 and one way in which it is different. [2]

Similarity:

The change in temperature stayed relatively constant before increasing exponentially/sharply ;

Difference:

there are more variations/fluctuations in the changes in temperature and less variations/fluctuations in atmospheric carbon dioxide concentrations ;

- (ii) Describe one human activity that has contributed to the enhanced greenhouse effect. [1]
- Burning of fossil fuels due to increasing energy usage releases large amounts of stored carbon into the atmosphere as carbon dioxide (CO₂) and is the major source of CO₂ emission ;**
 - Deforestation causes a net reduction in carbon storage as forested areas act as carbon sinks and results in CO₂ emission when forests are burnt, increasing the level of CO₂ in the atmosphere ;**

[Total: 6]

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